

ASSIGNMENT 12

ROLL NO:CSB24083

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from abc import ABC, abstractmethod
import json

# ===== PAYMENT =====
class PaymentMethod(ABC):

    @abstractmethod
    def pay(self, amount):
        pass

class CreditCard(PaymentMethod):

    def pay(self, amount):
        print(f"Paid Rs {amount} using Credit Card")

class UPI(PaymentMethod):

    def pay(self, amount):
        print(f"Paid Rs {amount} using UPI")

class WalletPayment(PaymentMethod):

    def pay(self, amount):
        print(f"Paid Rs {amount} using Wallet")
```

```
# ===== NOTIFICATION =====
class NotificationChannel(ABC):

    @abstractmethod
    def send_notification(self, message):
        pass

class Email(NotificationChannel):

    def send_notification(self, message):
        print(f"{message} sent through Email")

class SMS(NotificationChannel):

    def send_notification(self, message):
        print(f"{message} sent through SMS")

class PushNotification(NotificationChannel):

    def send_notification(self, message):
        print(f"{message} sent through Push Notification")

# ===== ORDER =====
class Order(ABC):

    def __init__(self, order_id, amount):
        self.order_id = order_id
        self.amount = amount

    @abstractmethod
    def get_total(self):
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        pass

class RegularOrder(Order):

    def get_total(self):
        return self.amount

class DiscountedOrder(Order):

    def get_total(self):
        return self.amount * 0.9

class PriorityOrder(Order):

    def get_total(self):
        return self.amount + 50

# ===== STORAGE =====
class StorageMechanism(ABC):

    @abstractmethod
    def save(self, order_data):
        pass

class JSONFileStorage(StorageMechanism):

    def save(self, order_data):

        with open("orders.json", "a") as file:
            json.dump(order_data, file)
            file.write("\n")
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        print("Order saved to JSON file")

# ===== ORDER SERVICE =====
class OrderService:

    def __init__(self,
                  payment_method: PaymentMethod,
                  notification: NotificationChannel,
                  storage: StorageMechanism):

        self.payment_method = payment_method
        self.notification = notification
        self.storage = storage

    def place_order(self, order: Order):

        total = order.get_total()

        # Payment
        self.payment_method.pay(total)

        # Prepare order data
        order_data = {
            "Order ID": order.order_id,
            "Original Amount": order.amount,
            "Final Amount": total,
            "Order Type": type(order).__name__,
            "Payment Method": type(self.payment_method).__name__
        }

        # Save data
        self.storage.save(order_data)

        # Notification
        self.notification.send_notification(
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        f"Order {order.order_id} placed successfully"
    )

    print("Order processing completed.\n")

# ===== MAIN PROGRAM =====
if __name__ == "__main__":

    # User Input
    order_id = int(input("Enter Order ID: "))
    amount = float(input("Enter Order Amount: "))

    print("\nSelect Order Type")
    print("1. Regular Order")
    print("2. Discounted Order")
    print("3. Priority Order")

    order_choice = int(input("Enter choice: "))

    # Order Selection
    if order_choice == 1:
        order = RegularOrder(order_id, amount)

    elif order_choice == 2:
        order = DiscountedOrder(order_id, amount)

    else:
        order = PriorityOrder(order_id, amount)

    # Payment Selection
    print("\nSelect Payment Method")
    print("1. Credit Card")
    print("2. UPI")
    print("3. Wallet")

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payment_choice = int(input("Enter choice: "))

if payment_choice == 1:
    payment = CreditCard()

elif payment_choice == 2:
    payment = UPI()

else:
    payment = WalletPayment()

# Notification Selection
print("\nSelect Notification Method")
print("1. Email")
print("2. SMS")
print("3. Push Notification")

notification_choice = int(input("Enter choice: "))

if notification_choice == 1:
    notification = Email()

elif notification_choice == 2:
    notification = SMS()

else:
    notification = PushNotification()

# Storage
storage = JSONFileStorage()

# Service
service = OrderService(payment, notification, storage)

# Place Order
service.place_order(order)
```

OUTPUT:

```
{ } orders.json > ...  
1  { "Order ID": 101, "Original Amount": 1000.0, "Final Amount": 900.0, "Order Type": "DiscountedOrder", "Payment Method": "UPI" }  
2  { "Order ID": 102, "Original Amount": 500.0, "Final Amount": 550.0, "Order Type": "PriorityOrder", "Payment Method": "WalletPayment" }  
3  { "Order ID": 103, "Original Amount": 700.0, "Final Amount": 700.0, "Order Type": "RegularOrder", "Payment Method": "CreditCard" }  
4
```

Enter Order ID: 103

Enter Order Amount: 700

Select Order Type

1. Regular Order
2. Discounted Order
3. Priority Order

Enter choice: 1

Select Payment Method

1. Credit Card
2. UPI
3. Wallet

Enter choice: 1

Select Notification Method

2. SMS
3. Push Notification

Enter choice: 3

Paid Rs 700.0 using Credit Card

Order saved to JSON file

Order 103 placed successfully sent through Push Notification

Order processing completed.